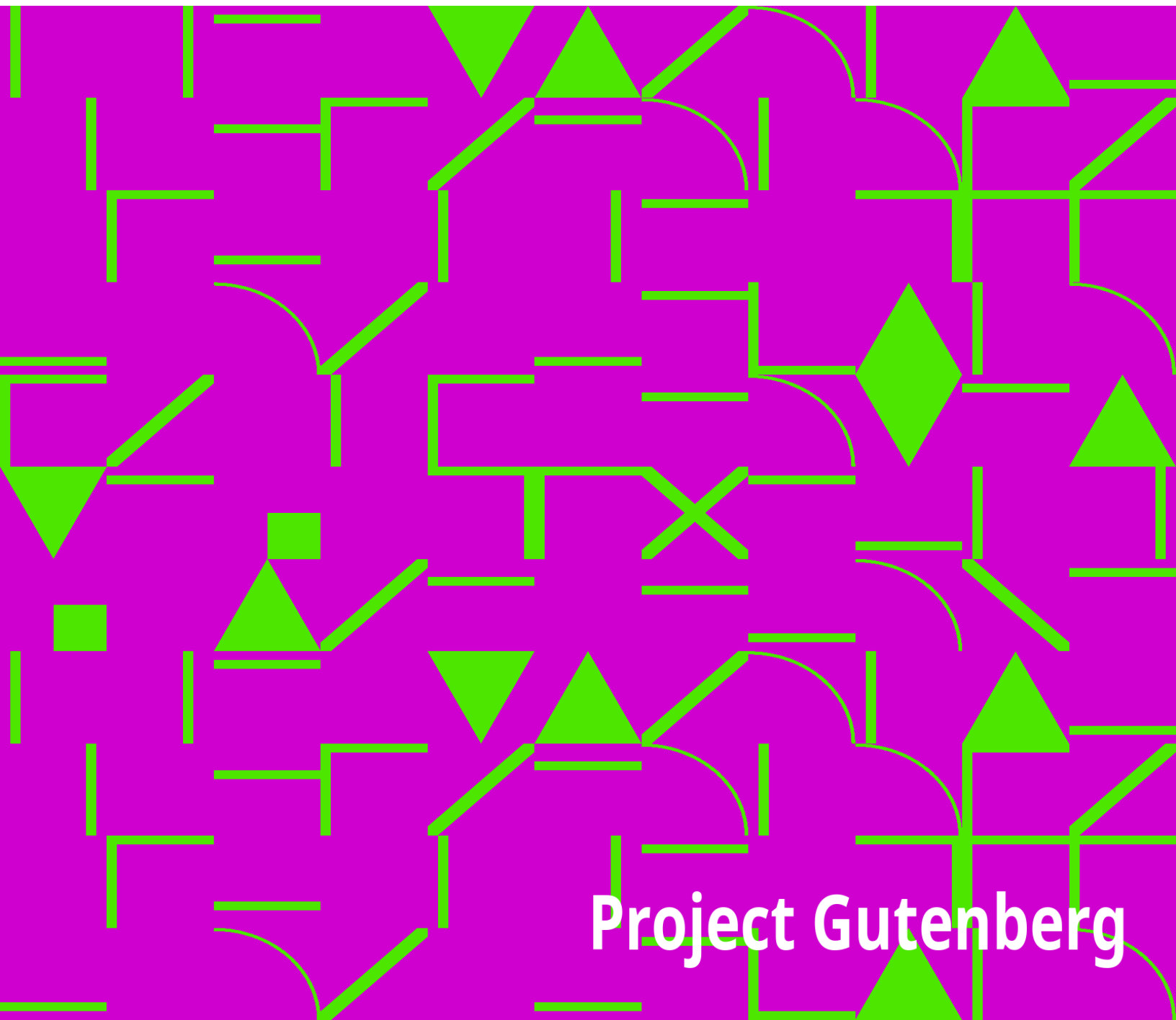


# The Sanitary Condition of the Poor in Relation to Disease, Poverty, and Crime

Benson Baker



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CONDITION OF THE POOR IN RELATION TO DISEASE, POVERTY,  
AND CRIME \*\*\*

# THE SANITARY CONDITION OF THE POOR

IN RELATION TO  
DISEASE, POVERTY, AND CRIME  
WITH AN APPENDIX  
ON THE CONTROL AND PREVENTION OF  
INFECTIOUS DISEASES.

BY  
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SAINT MARYLEBONE, MEMBER OF THE NATIONAL ASSOCIATION  
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IN MIDWIFERY, FELLOW OF THE OBSTETRICAL  
SOCIETY OF LONDON, L.S.A., F.A.S.L.

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## PREFACE.

The following pages having already appeared in the form of letters in the *Marylebone Mercury*, any preface on my part might be deemed unnecessary. Having been repeatedly asked to collect and publish them, I intended to revise if not rewrite the whole, but my professional duties have deprived me of the time, so at once I place them in the hands of my indulgent readers. This pamphlet does not pretend to be a treatise on Hygiène, but simply to hint at some of the evils which arise from the neglect of sanitary regulations, also to suggest some simple remedies, and further to try and induce men to regard health as one of the most valuable and sacred gifts of God.

BENSON BAKER.

94, *Lisson Grove*, N.W.

**THE  
SANITARY CONDITION OF THE POOR,  
IN RELATION TO  
DISEASE, POVERTY, AND CRIME.**

THE relation of the rich to the poor, and the poor to the rich, has always been a subject of interest to the philosopher, moralist, economist, and philanthropist. This relation was, in feudal times, clearly enunciated, and as vigorously acted upon. The dark days and long evenings that witnessed the tolling of the curfew bell have passed away. Since those days we have made advances in social and political freedom, and class distinctions have become less obvious. Freedom of thought has developed a greater equality of social rights; but are the poor really any better off now than they were in the days when William hunted in the New Forest? I confess that many will reply in the affirmative; but when I see around me on every side so much disease, poverty, utter wretchedness, and crime, I conceive it difficult for the condition of man to be worse than that in which thousands in this metropolis exist. Look at the condition of the overcrowded dwellings of the poor; they are not *homes*. The word home in its full and happy significance and association is unknown to them. The proud boast of every Englishman, that his house is his castle, is a mere fiction. "There is no place like home, be it ever so humble," is a truth that meets with the faintest response from thousands of miserable tenants who live in yet more miserable tenements.

That the condition of the dwellings of the poor does not free them from the obligation to observe personal cleanliness is perfectly true, neither does the neglect of personal cleanliness on their part justify the landlord in neglecting the sanitary condition of his houses. It is to be regretted that more liberal and enlightened measures are not adopted. I feel confident that if the dwellings of the poor were made healthy and comfortable, there

would be a marked improvement in the status of the poor. With improved health, cleanliness, and comfort we should obtain improved industry and prosperity; and this would not be confined exclusively to the poor, but its influence would be felt through every grade of society. It would tend to lessen the selfishness of the prosperous, and the suspicious jealousy and impostures of the unfortunate. We are apt to forget that:—

“Is there for honest poverty  
That hangs his head, and a’ that?  
The coward slave, we pass him by,  
And dare be poor, for a’ that.  
The rank is but the guinea stamp,  
The man’s the gowd, for a’ that.”

Man is to a very great extent the creature of circumstances, and is greatly influenced by external objects. That which we like we become like. The fine arts cultivate the æsthetic part of a man’s nature; painting, sculpture, and poetry develop a man’s imagination. One man by a constant contemplation of the beautiful acquires the habit of admiring the beautiful; another man by the constant association with dirt becomes dirty. Repeated acts of thought grow into action, and actions repeated grow into habits, and habits form the character of the man: hence the importance of a cleanly, comfortable, and healthy home. A family that lives in a dirty room, with nothing but dirty associations, must grow up dirty. Thoughts grow in us as grain in wood. The mental qualities of the poor are undeveloped, and especially the moral force of character which is requisite to raise them above their circumstances is hardly to be looked for. If the poor could be accommodated with healthy dwellings they might reasonably be expected to better their condition; but what man can wage a successful war against disease and poverty combined? Unhealthy dwellings not only keep them poor, but foster a reckless indifference as to how things go with them. This apathetic condition is the worst that can befall either the individual or society. Idleness, dirt, poverty, disease, and crime are intimate companions, though not inseparable. The logic of the indifferent poor amounts to this: that those who work must pay for those that don’t; and hence arises one of the great difficulties in dealing with and relieving the poor. There are many limited companies formed to carry out various commercial enterprises; but

if a limited company could be formed to erect suitable dwellings for the poor, at such a rental as they now pay, it would confer an unspeakable benefit on the poor, benefit society generally, and pay the really patriotic shareholders an equitable percentage. This I merely throw out as a suggestion, but I am convinced if carried out, it would be one of the best means of effectually helping the poor to help themselves. I conceive that whatever tends to smooth the rugged surface of life, and to soften its harsher features, and to multiply the sources of human health and enjoyment, must be held worthy of notice; and in so far as it tends to improve the condition of the poor, and foster in them provident and careful habits, just so far ought it to receive our generous sympathy and support. That this may not be mistaken to be merely philanthropic statement, but that it is based upon a principle of true social economy, I quote the following from Smith's *Wealth of Nations*:

“Servants, labourers, and workmen of different kinds, make up by far the greater portion of every great political society. But what improves the circumstances of the greater part can never be regarded as an inconveniency to the whole. No society can surely be flourishing and happy of which the far greater part of the members are poor and miserable. It is but equity—besides that they who feed, clothe, and lodge, the whole body of the people should have such a share, as to be themselves tolerably well fed, clothed, and lodged. The liberal reward of labour, as it is the effect of increasing wealth, so it is the cause of increasing population. To complain of it is to lament over the necessary effect and cause of the greatest public prosperity. It deserves to be remarked perhaps, that it is in the progressive state, while the society is advancing to the further acquisition, rather than when it has acquired its complement of riches, that the condition of the labouring poor, of the great body of the people seems to be the happiest and most comfortable. It is hard in the stationary, and miserable in the declining state.”

In a visit I paid to a poor patient in Burne Street, I could not help being struck with the cold, damp, dirty condition of this “attic near the sky.” The cold wind pierced through every crevice. The rain had come in plentifully through the ceiling and soaked the bed. The walls were black and grimy;



the fire in the grate, like the life of my patient, feebly struggled for existence.

What dirt and neglect had done for the room—dirt, disease, and poverty had done for the inmate. I feel sure that had the dwelling been in a better condition my patient would have been in better health, and consequently stood better in every relation to society.

'Tis an old proverb that “prevention is better than cure;” yet how few there are who practically believe it. The principle embodied in this proverb received the approval and sanction of past ages, but which it is most difficult to enforce at the present time; it may be explained by the fact, that men not being forewarned cannot be forearmed, and further, because another principle which is expressed in the words, “men think all men mortal but themselves,” lies deep in the human mind. The blessing of health is not valued till it is lost. Let the mysterious hand of disease touch us, and we throb with painful anguish. Then do we long most earnestly for our valued health and strength; we resolve that once well again we will adopt such regulations and precautions as shall ensure a continuance of the invaluable blessings of health: such is the reasoning of the sick and afflicted. To the invalid the quantity and quality of the air he breathes is of vital importance, but to the man in comparative health, it is a matter of indifference. The laws which regulate the diffusion of gases on which ventilation depends, or the action of pure or impure air on the animal economy, is to him a subject of indifference; it does not yet affect him, and he has enough to do without it. The subject of ventilation bears such an important relation to health, and health is one of the greatest blessings of life, that I conceive it to be the duty of all to be fully alive to its value. In these days when the schoolmaster is abroad, and the march of intellect so rapid, it is not a little remarkable that the knowledge of the laws of health, based upon scientific data, are so little known.

If therefore, some of the general laws which govern the preservation of health be simply stated and illustrated, it may be the beginning of much good, by teaching or inciting men to become the guardians of their own health. One of the primary laws of health is to have a good supply of fresh air. The voice of nature cries aloud for fresh air, the cry comes to us alike from the newly born babe in the cellar, and the aged sufferer worn out in

life's troubles in the garret. It is the cry of the weary and worn; their thin blood flows feebly, and their life is ebbing fast as they cry, "Oh, for one short hour to breathe the breath of the cowslip sweet and to feel as I used to feel." This is all that is craved ere the weary one welcomes the stifling hand of death. Such is the cry of thousands in our city. Who wonders at it? Not those who are practically acquainted with the dwellings of the poor. For those who are not, I shall in general terms indicate some of the features of the *homes* in my district. It is not my object to raise the veil of poverty to satisfy idle curiosity, or to awaken a morbid philanthropy. Scenes which are photographed on my mind, I shall from feelings of delicacy alike for the subject and the reader omit. As parochial medical officer for Christ Church district, it is very often my duty to visit East Cottages, and Little Church Street, and I am daily called upon to witness the suffering and sickness in crowded rooms, which is fostered by bad ventilation and insufficient drainage; here small-pox and typhus fever are frequent visitors. It is not an uncommon thing for a family of eight or ten to occupy one small room. In this single apartment, men women and children of all ages eat, drink, and sleep; in such a place as this, and under such circumstances, where little air enters, and less light, can it be a matter of surprise that sickness and death are frequent visitors. Death is so frequent a visitor, that to those who witness his work he has lost all his terrors; yea, he is rather a friendly visitor when he calls the younger members home. Think, reader, of such a *home* as this in the day of sickness, in the hour of nature's woe, in the gloomy night of death. Picture children of tender years becoming familiar with these scenes, and can it reasonably be expected that they should grow up healthy either in mind or body. I could not but allude to the number of occupants in one small apartment, as it bears directly on the amount of air supplied to each one, and the necessity for free ventilation; overcrowding also tends very materially to demoralise the rising generation. In a word, it degenerates both body and mind. At an early age they sow the seeds of consumption, bronchitis, rheumatism, and a whole list of complaints, which may be traced more or less to the violation of the ordinary sanitary laws of health. The youth of our cities before they are young grow old, before the bud blooms the blossom is blasted; as flowers without the light of heaven, they are blanched, withered, and die. These sickly ones crawl into their homes, such as they are, from which the light of day is almost excluded, and with it the light of hope. Under these circumstances the parish doctor is

sent for, but too often it is his lot only to be a passive witness of the irreparable mischief which has been done. I have a vivid and painful recollection of a visit I paid to Devonshire-place. The room was small, dark, dirty, and gloomy. There were but few squares of glass left in the frame, through which the yellow light feebly struggled to enter; the chimney was stopped up, the air stagnant and very oppressive, even for the few minutes I was present, yet there were no less than five people sleeping in this apartment, two children ill with scarlet fever, and one resting in death. The ventilation was so bad, that the recovery of my little patients was doubtful, and the deterioration of the health of their parents certain. In cases of this kind how important it is for every one to know, that by excluding the breath of heaven, we are extinguishing the breath of life. It is of the highest importance that we should have fresh air in health, but it is imperatively necessary in disease. Then more than ever does nature feel the necessity for fresh air. A lowered vitality like a fire that is dying out, requires a larger supply of atmospheric air, but alas! how often is this innate craving for fresh air stifled by prejudice and ignorance. Close rooms are rendered closer by chimneys being blocked up, and every crook and crevice being stuffed up by rags and paper, as if the great object to be attained was effectually to prevent one breathing. All this is done from the purest motive, and with the best intention; it is done to prevent the patient taking cold. So the patient has to struggle against sickness, poverty, and impure air.

Those who are in health, though they may sleep in overcrowded rooms, have the advantage of being out in the open air some part of the day, and this in some measure compensates for their close confinement. The open air quickens the pulse and restores the vitality of the various functions of the body. Fortunate are those whose employment is out of doors. This obtains rather with the male than the female poor. They too often leave a closely confined and overcrowded bed-room to go into a still worse work-room, there to work *twelve* or *fourteen* hours under flaring gas-lights in a heated and impure atmosphere. Those who live like this succumb at an early period of life to disease:—

“With fingers weary and worn,  
With eye-lids heavy and red,”

they rush out of their overcrowded work-rooms and seek to palliate their physical sufferings at the gin-shop.

The air which surrounds us on every side, though invisible, is not inactive. To the casual observer its existence might be doubted if it did not manifest itself in the gentle breezes which bear the faded leaves of the widowed trees to their resting place; or in the wintry blast which uproots the sturdy oak of the forest, or lashes the white-crested billows of the ocean into a storm. In this way does the air we breathe make itself known to us. The composition of atmospheric air, the laws which govern it, as illustrating the theory of ventilation, the action of air on the animal economy, on the vegetable and mineral kingdoms, we must pass by. Suffice it to say that atmospheric air is a compound; that oxygen is the life-sustaining element; and that carbonic acid gas has an opposite effect; also the nitrogen; and that when we breathe we take in oxygen and give out carbonic acid gas. There is a law which governs cases called gaseous diffusion, by which gases pass equally into one another, and thus become equally diffused. If it were not for the operation of this law, there would be no variation of climate, no winds to waft the mariner over the briny deep, no provision for the scattering of the sun's golden rays of light. The beauty of a summer's sky would have been lost. No clouds would have hung around the portals of the setting sun tinged with crimson and with gold. No verdure in the meadow would have greeted our eye. No ark of hope would have spanned the clear vault of heaven. But, by the combination of oxygen and hydrogen, and the laws which govern them, provision is made for all these blessings. Light and heat are freely scattered. Animal and vegetable life rejoice. The priests of the morning chant their matins when the sun arises, and the flowers scatter their evening incense when the sun descends to glorify the western world.

In a sanitary point of view the law of gaseous diffusion is of importance, because noxious gases are thereby diluted by the atmosphere, and are thus rendered comparatively harmless; hence, by free ventilation, we may do much to lessen the predisposition to epidemics, and the virulence of poisonous malaria.

The labouring man, above all others, should rise in the morning refreshed by nature's sweet restorer—balmy sleep. But, by neglecting ventilation, it is not so. Drowsiness, sickness, headache, and languor are his lot; and in

order to bring himself up to the mark, he is obliged to have recourse to artificial stimulants. Scavengers are necessary to maintain the comfort and cleanliness of our streets, so oxygen is necessary to maintain the health and comfort of our bodies. By depriving our bodies of pure air, we deprive them of the power to throw off infectious diseases. This may simply be illustrated by placing a lighted candle under a shade when a limited amount of air only can be introduced. If not supplied the candle will go on burning for a while, giving out less light and less heat; but when the oxygen is exhausted, the candle will die out. Thus it is with man. The candle, when deprived of pure air, gave evidence of less light, so to man under the same conditions would give evidence of less life. A lower vitality is equivalent to a closer approach to death. Day is opposed to night. Life to death. But even as between day and night there is twilight, so in our experience there is the twilight of disease. Thousands in this metropolis, through the neglect of the laws of health, live in the twilight of disease, rather than in the noon-day brightness of health.

I have endeavoured to show that fresh air is a necessary of life just as much as food. I have indicated its importance in relation to health and how imperatively necessary it was in sickness to aid the body to overcome disease. If overcrowding in small dwellings and its consequent ill effects upon health, morality, and industry be a necessary social evil, it is evidently our duty to lessen that evil, if we cannot altogether remove it. This, in some measure, may be accomplished by free ventilation and paying due regard to the ordinary sanitary laws of health. If we turn a deaf ear to the warning voices of conscience and reason, and go on living and breaking the laws of health, we must expect to pay the penalty for breaking those laws. If men habitually sleep in an atmosphere of carbonic acid gas and fetid exhalations, they must of necessity exhaust the vitality of their bodies.

The defective sanitary condition of hundreds of houses in this district render it imperative that something should not only be said, but done. But the question arises—who is to do it? I take it that the duty lies not alone with either the tenant, or landlord, or parochial authorities. It is a work of co-operation in which there is work for each; and each must do his work. In endeavouring to urge this sanitary reformation it may be Utopian to expect a perfect result; yet I am convinced that much can be done to render even the present wretched dwellings of the poor much more healthy than they are

at present. I have frequently in my visits to Charles-street, Stephen-street, Stephen's-court, and, in fact, in the houses contiguous, met, on going upstairs in the morning, a hot, fetid air. On going into the rooms I have observed that the walls were damp with exhalations caused by so many herding together. I have naturally requested the windows to be opened, in order to let out the hot and impure air, and, to my dismay, I have almost invariably found that the upper sash was a fixture. This state of things is to be deplored, inasmuch as it prevents efficient ventilation and is not only injurious to health but beyond the means of poor lodgers to remedy. I would venture to suggest the propriety of landlords seeing that all windows be made to open at the top, or a square of glass be removed and a perforated tin plate be inserted in its place, or better still, a window ventilator. A brick removed out of the chimney, close to the ceiling, and a valve fitted so as to allow the escape of foul air up the flue, is a good means of ventilation. There are many ways of ventilating an apartment; and it matters little how it is done if it be done efficiently. If any one of the above methods be adopted I feel sure it would be a great boon to many poor.

There is one subject that bears a most important relation to the health and comfort of those over-crowded dwellings, and that is the supply of water. This is anything but adequate to the requirements of the many people that are sheltered under one roof in my district. The tubs and cisterns are in many cases only sufficient to supply one family, but in the houses where the poor live, each room is occupied by one or two families at the least; the consequence is, that there is a great demand for water, and not being sufficient, impurities of all kinds abound; personal cleanliness is neglected to a frightful extent, washing the skin is looked upon as a terrible innovation, and the result is often dreaded. To suggest a bath is often looked upon with as great a horror as a surgical operation. If there were a more plentiful supply of water, there would not be the excuse for such a neglect of washing and cleansing the dwellings of the poor. Personal cleanliness and the washing out of rooms, &c., is the duty of the tenant quite as much as paying the rent. It would be well if cleanliness were made a condition of tenancy,—if the rent collector insisted on the cleanliness of the rooms with the same firmness that he insists on the punctual payment of the rent, yea, rather allow a week's rent to stand over than a dirty room to go unwashed.

The sewerage to the houses is not adequate to carry off the enormous waste of fifty people. These houses were only built for the accommodation of fifteen to eighteen people. If there be not a sufficient supply of water to flush the drains at least once a-day, they become clogged up and generate poisonous gases, which enter the houses and bring on a lowered state of health; skin diseases, scabies, typhus, and choleraic diarrhœa are the results. That this is not an overstatement may be shown by the fact, that outbreaks of choleraic diarrhœa have over and over again been suddenly stopped by a heavy fall of rain. This beneficial result has come about by the rain flushing the sewers and clearing off the morbid poisons.

There are other nuisances which contribute to contaminate the air. I have, in visits from house to house, noticed that the dust heaps are allowed to accumulate to a frightful extent, they are neither more nor less than one vast mass of decayed animal and vegetable matter, with a good proportion of the remains of decomposed fish. The consequence was that if windows were opened for the purpose of ventilation, a worse condition of the room was induced. Further, I cannot help remarking that the sanitary condition of the streets and courts in my district, especially part of Bell-street, Charles-street, George-street, part of Devonshire-street, is far from satisfactory. Those that live in the cellars in these streets are constantly inhaling poisonous exhalations arising from putrid animal and vegetable matter, and I am sure the stench which arises from stinking muscles and cockle shells, must be injurious to health. I would ask upon what principle do the authorities do their work? Broad streets, squares, and open thoroughfares appear to me to be oftener watered and swept than these out-of-the-way courts, alleys, and narrow streets, which are overcrowded with a teeming dirty population. Surely these are just the places that require the greatest care, for these are the places which, if they do not breed an epidemic, would readily foster it. The slums of a city ought to be made as healthy as possible, not only for the good of those who live in the slums, but for the good of the community at large. Society is like the body; one member cannot be affected without the whole body suffering. It is not only right, but profitable to make the dwellings of the poor healthy, and to surround them by all the protection that preventive measures can do. Sickness is one of the worst calamities that can befall the poor labouring man. When laid low, all his resources are cut off. His family, and so must he, go without the

necessaries of life, or run up a heavy score at the small huxter's shop. His rent cannot be paid; and should his illness last long, both he and his family become a burden to the ratepayers; and should he recover his health, it is weeks, aye, months before he can recover his position and stand free from debt. Now, suppose that this illness was brought on by the defective sanitary condition of his dwelling, and his own neglect of proper attention to cleanliness, why it is at once evident that it is the interest of the landlord to see that his houses are healthy, for a dead man pays no debts, and a sick man pays no rent; further, it is the interest of the tenant to observe the laws of health. What pen can describe the racking torture of disease and poverty combined? I have visited houses where the head of the family has been thrown of work, by ill-health. I have witnessed the gradual departure of articles of ornament from his little room, soon to be followed by other pieces of furniture, until even the necessary utensils of civilised life were gone. The bedstead and bedding have been put away, and a sack of shavings supplied its place. The room has been literally stripped. The children have been equally deprived of their clothes, and were huddled up in a corner in quite as primitive a fashion as their forefathers the ancient Britons. The mother went out washing for two or three days a week; what she earned they all lived on. It may be asked why did he not go to the workhouse? I reply that the deserving poor, the earnest, striving poor, have the greatest possible objection to go to the house. They would rather die in want and wretchedness than become an inmate of the house. Whether this dislike be well founded or no, it is not my object now to inquire, but I must state that there are some clamorous for admission. From what has been said, I think it may fairly be inferred that if a liberal and just regard be paid to the sanitary condition of the dwellings of the poor it would, to look at it from the £ s. d. point of view, be a benefit to tenant, landlord, and ratepayer. It would also tend very materially to improve the physical, moral and mental condition of the poor; it would make them healthier men, happier men. It would materially better fit them for all the social conditions of life.

I know of no subject in a sanitary point of view that demands a more careful consideration than the possibility of a choleraic epidemic next spring. I wish not to alarm, but to try and urge people to be on their guard so as effectually to lessen if not prevent the dire effects of a cholera visitation.



Should cholera visit this part of the metropolis—and we have no guarantee that it will not—it is our duty to be prepared to deal promptly and effectually with it.

The news has been already flashed through the land that it has made its appearance. Although confined at present to two places, this has made many a stout heart tremble. But this is not a time for fear. “Let us, then, be up and doing with a heart for any fate,” and seek by an earnest endeavour to render our homes untenable to this terrible pestilence. Every man this day must do his duty. These are stirring words, and they call us to action, for a deadlier foe now threatens the happiness of our sea-girt island than that which Nelson fought. Cholera, an unseen foe, is nevertheless a real one. Many a hero that has braved a hundred fights has blanched before this deadlier foe. Like a mighty conqueror and destroyer, it has marched onward as regardless of the ermine of the judge as the swaddling clothes of the infant. In vain has the mother wrung her hands in passionate anguish over the death of her little one; husband has mourned for wife, and wife for husband; children have been left fatherless; and fathers have been left childless; nothing but darkness, blackness, death, and desolation, have marked the route of this potent enemy to man.

One of the best preventive measures is to have a plentiful supply of pure water. We can then keep our sewers flushed—our homes clean, and so render our hearths and homes comparatively safe from the devastating pest of cholera.

There is nothing in nature that possesses more interest than water. It is composed of oxygen and hydrogen. This may be easily demonstrated by throwing a piece of potassium on water, when it will instantly take fire: because of the rapid affinity of the oxygen of the water for the metal potassium the hydrogen is liberated, which burns during the decomposition which takes place. This experiment, which is equally simple and beautiful, shows the composition of water. The combinations into which water enters in the animal and vegetable kingdoms are as various and as beautiful as the magic forms of the kaleidoscope. Neither animals nor plants can live without water. It is, therefore, a necessary of life. On the purity of water, in a great measure, our health depends. Various are the sources of water, and equally various the degrees of the purity or impurity of water: for this

reason—that water is the great solvent in nature. The first great source of our water supply is from the ocean, which receives the water from all the sparkling rills that murmur as they leap down the mountain side, which unite and form rivulets. These rivulets unite again, and form the mighty flowing rivers with majestic water-falls, cataracts, and cascades; and these flow on till they empty themselves into the ocean.

Reader, did you ever stand on the sea beach shore and see day go out in a flood of glory—the bosom of the ocean blush as the monarch of day smiled his parting beams of light—tinging the clouds with golden hue and shedding a light on all creation and making it beautiful? These clouds were not alone created for the poet or the painter, nor to give scenic beauty to the landscape; but to supply the wants of the flowers of the valley, the beasts of the forest, and to be a source of health, power, and wealth to man. The heating rays from the sun so act upon the water as to cause a mist like a gossamer veil to arise. This is tossed about in the atmosphere by the various currents of winds which set in from the Equator to the Arctic and Antarctic regions. Thus is it that rain-bearing clouds are distributed over the land. The vapour contained in these clouds is condensed by the various degrees of temperature through which they are wafted, and form either snow or rain. This sinks down into the earth, and forms springs; or the snow, melting on the mountains, runs down and forms rivulets: and hence our chief water supply.

“Look around on this world, it is sweet, it is fair,  
There is light in the sky, there is *life* in its air,  
Sublimity breathes from the forms of its hills,  
And beauty winds on with its rivers and rills;  
The dew as with diamonds its mead hath bespread,  
From its groves are a thousand wild melodies sent;  
While flowers of each tint are by morning imparted,  
Oh! why is there woe in so lovely a world.”

Rain water, when collected in the country, is doubtless the purest and best adapted for ordinary household purposes, but in London and large cities it contains many impurities; when collected in our tubs or cisterns, it is impregnated with all the dust, dirt, and smoke that hangs more or less like a funeral pall over smoky cities; the air contains more or less sulphurous acid,

and the carbon that escapes from our chimneys and is deposited on the tops of our houses, hence the sooty taste that soft or rain water has.

Well or spring water always partakes more or less of the character of the ground or the strata through which it passes, hence the explanation of the great differences which exist. Water that passes through chalk is characterised by its hardness, because it holds a large quantity of carbonate of lime in solution. Thus it is with chalybeate springs, which contain a large proportion of the salts of iron, and others, that contain sulphate of magnesia; these instances illustrate the fact that water is a great solvent, and partakes of the character of the materials through which it passes, and hence the differences which we find existing in the water we derive from wells. There are not many public pumps in my district, and fortunately they are not in the vicinity of churchyards, nor in communication with cesspools. There is not therefore much danger of wholesale poisoning by drinking water from these sources. The water, as supplied by the West London and Grand Junction Companies, is, I believe, of good quality; for from the samples I have examined, and the reports furnished by Dr. Whitmore, the water appears free from those organic impurities which are indications of danger, and are specially to be guarded against in times when cholera threatens us. The water, which in not a few of the houses in my district, is supplied by the water companies pure, is rendered impure in the tubs and cisterns in which it is stored. Water not only has the property of becoming impregnated with mineral substances, but also the property of absorbing gases. It is strange, but nevertheless true, that the worst place possible for tubs, cisterns, and water reservoirs, is selected, viz., next to or over water closets, or over dust bins; this obtains almost universally in my district. The consequence is, in the one case feculent exhalations arise, and are absorbed by the water; and in the other, exhalations from the decomposed and decomposing vegetable and animal refuse; further, the tubs and cisterns in many cases appear to have been neglected for a long time. The lids do not fit, consequently these cisterns are receptacles for all kinds of filth. Those who live in the upper rooms are not over-scrupulous as to how, when, or where, they dispose of their waste matters, but it often finds a lodgement on the outbuildings, and not seldom is washed more or less into the cisterns. These cisterns, like the people and the rooms they inhabit, are not characterised by cleanliness. If I might judge from appearances, I should

conclude that they are seldom washed out. If I ask when the tub or cistern was last cleaned out, nobody seems to know; for, what is everybody's duty is nobody's work.

It is a well-known fact, that when animal and feculent matters are suspended in water, and that water is used for drinking purposes, that it has produced diarrhœa. In water that is contaminated with sewerage the diarrhœa is accompanied with choleric symptoms, such as purging, vomiting, cramp, and even some loss of heat. This should be a warning not to allow impurities of any kind to abound in our drinking water. Fetid gases absorbed by water render the water bad; not that the sulphuretted hydrogen in itself is the thing to be dreaded, but because its presence fosters the growth of vegetable and animal products that are highly detrimental to health, and are, I believe, predisposing causes, if not virtually the agents in choleraic diarrhœa. It is in water under these conditions that plants and animals abound. They are minute, microscopic, I admit, but on that account none the less dangerous. The living plants and animals are not to be regarded with so much horror as the dead, because they rapidly undergo decomposition; and, while in this state, if taken into the system, act as ferments disturbing the normal functions of the body, and often giving rise to diarrhœa. This is not a mere assertion or hypothesis, but is supported by well-observed cases by some of the most competent men in the medical profession. In support of the above, I quote the following words from Dr. Parkes, professor of military hygiène. He says:—"An epidemic of diarrhoea in a community is almost always owing to either impure air, *impure water*, or bad food." The practical deduction is, that if we would maintain the health of a parish and prevent cholera, diarrhœa, typhus, or any other epidemic, we must improve our sanitary condition. Pure water lies at the foundation of all sanitary improvements. Let it be well known that impure water is to be dreaded, and the fear of an epidemic may prove a healthy stimulus to increased activity and cleanliness in connection with a subject of such vital importance. Whether cholera visits us or not, we shall be all the better for having paid proper attention to our sanitary arrangements. I would suggest the propriety of having all tubs, cisterns and reservoirs cleaned out; all waste matters removed from the roofs of outhouses; all back premises thoroughly cleaned. I think from what I have said, this can hardly be regarded an unnecessary work; but should this be

deemed impracticable—though I cannot see how it can be—I would suggest the desirability of all water used for drinking and culinary purposes being filtered; that is, where there is any suspicion of decomposed organic matter in the cistern, tub, &c. Any working man with ordinary ingenuity might contrive to filter his water through a little sand and vegetable charcoal. I know of no filters that could be purchased for 1s. 6d. or 2s., which I presume is about the extent that many of the poor in my district could go to. A rough-and-ready filter might be made out of a large flower-pot, the hole at the bottom being plugged up with sponge, and then a layer of charcoal and another layer of sand. The water would percolate through these substances, and be as clean and pure as the water from the mountain rill. The flowerpot might be suspended over another vessel, which would receive the pure water, or it might be so placed as to stand on two pieces of wood over another vessel. There are many ways of filtering water, but it is only my object to show how simple, inexpensive it is, and that it is within the means of the poorest to protect themselves from the disastrous effects of impure water, at a little cost of labour and at a still less cost of money.

There can be no doubt that water is not appreciated as a sanitary agent. This may arise either from ignorance or indifference; but, after what has been said, I trust that the apathetic neglect of the primary laws of health, and that blind confidence in an overruling Providence, that in the end all will come well—may vanish as the baseless fabric of a vision. It is in vain to call upon the local authorities to supply a larger quantity of water, if we either from ignorance or indolence neglect to use it. It is in vain to call upon landlords to build houses properly drained and ventilated if those who take them are ignorant and indifferent to the value of fresh air and good drainage. Dirt is allowed to begrime the walls, so as to render it difficult to tell what the original colour was; in a word, many of the rooms in George-street, Charles-street, Little Bell-street, Little Church-street, East-cottages, and many others in my district, are so neglected by the tenants as to become neither more nor less than fever nests. Soap and water and white lime, to judge from the condition of very many of the rooms in my district, might be as expensive a luxury as salt in Abyssinia. Every traveller that visits tropical countries is sure to complain of the annoyance and inconvenience of mosquitoes, and I am equally sure that every visitor to the dwellings of the poor in the locality I have indicated might reasonably complain of the

visitors, who, on account of the dirt, have taken up a permanent residence, but whose name it is not deemed polite to mention, though as far as number is concerned they might be justly termed legion.

Almost without exception the poor bear the marks on their own body of their careless indifference to the laws of health and cleanliness. I do not believe that the poor are too bad to be mended, nor that they are too dirty to be taught the necessity and desirability of cleanliness. Those who suffer thus from sanitary mismanagement are not to be improved by calling them filthy, incorrigible, and worthless, but by kindly pointing out the evils which arise from their own carelessness, and suggesting remedies which they themselves can carry into operation. The very worst have some good in them, and by appealing to that good in a kindly, honest spirit we may arouse that laudable ambition of self-respect and self-help, which will do more to improve the sanitary and social condition of the needy poor than could be accomplished by extraneous help, if it did not secure the hearty sympathy and co-operation of the poor. Until the poor take an interest in their homes and feel a pride in preserving them clean, healthy, and comfortable, we shall always be hearing of and seeing the homes of the poor in a dilapidated condition. Flooring and doors will be used as firewood—sewers will be blocked up by materials that ought never to have been thrown down—dirt, filth, and all manner of rubbish will be allowed to accumulate—how is this to be remedied? To remedy this evil, mere obstructiveness and selfishness must get out of the way. The homes of the poor must be cleansed, repaired, and made comfortable; and then if cleanliness were made a condition of tenancy as I have had occasion before to suggest, it would go a long way to keep the property in repair. I feel confident that where tenants keep their rooms clean they acquire an interest in them, and do not destroy nor wilfully damage them. Further, every effort ought to be made to instruct the poor in the knowledge of the laws of health. Let it be clearly shown to them that dirt, disease, and poverty, are intimate companions, and that if they would escape from the bitter experience of poverty, they must each sweep before their own doors; then a future, gilded with the promise of a joyous hope, shall open up before their view. Dirt with all its disgusting associations shall be supplanted by cleanliness with all its cheering comforts; disease manifesting itself in the emaciated body, the anxious brow, the glazed eye, and shrivelled face, shall

give place to joyous health, with firm elastic step, with ruddy face, bright sparkling eye, and the alacrity and pleasure with which labour and toil is welcomed. Poverty in all its intensified bitterness, when associated with disease shall give place to a comfortable competency. This and much more may justly be expected, if we sow aright the seeds of the science of life, and teach men the laws which govern their health and life. The proper study of mankind is man; a more comprehensive principle has seldom been enunciated. Man in all his relations is too vast a subject for us to contemplate, therefore we purpose the consideration of man in his social relations, in so far as these relations bear upon the health of the individual or the community of which we form a part. It would be well if we all took a deeper interest in the welfare of each other, if we felt that any disease, epidemic, or misfortune, that affects any member of the community, must by the common law of relationship, affect us; the man who wilfully breaks the laws of health, not only injures himself and his next of kin, but he injures society, of which he is a member, by depriving society of the creative wealth of labour; hence we see that the individual welfare and the prosperity of the commonwealth, depends upon a right understanding of those great laws by which the Creator governs the health and life of man. I know of few places where personal cleanliness is at a lower standard than in many of the small streets in my district, therefore at the risk of being personal, I must press the subject of the importance of baths. All nations have valued baths, some have regarded ablutions as a necessary part of their religion, and have accordingly most faithfully carried into daily practice their ablutions. Others have regarded the bath as a luxury, and only to be indulged in occasionally. There are those who inhabit the far, far north, where mountains of ice wreathed in the scintillations of the northern lights tower in colossal grandeur. The cold is so intense, that animal and vegetable life struggle feebly to maintain an existence, nevertheless there man prizes his bath. There are nations basking beneath a tropical sun, where vegetation revels in luxurious richness, where birds with gorgeous plumage carol in the clear blue sky, and dolphins as they sport in the water sparkle and glitter like a thousand gems, and here man values the bath as one of the best gifts of the great Creator. Nor was ancient Greece or Rome backward in discovering the value of the bath as a sanitary and curative agent. They bear us ample testimony of the high value they put upon baths. In this short paper it would be an impossibility to discuss the various

kinds of baths, hot, shower, swimming, cold, Turkish, and vapour baths; neither can it be expected that I should indicate the relative value of each to particular conditions of health or disease, but we may rest assured that each kind of bath may, when rightly used, be regarded as a health preserving and health restoring agent.

In order fairly to estimate the value of baths, we ought to know something of the structure and functions of the skin. The skin is furnished with millions of little holes and tubes, these holes are called the pores of the skin, through which watery vapour or sweat is poured out; it is estimated that there are no less than 3,000 of the tubes and holes pouring out sweat in space not larger than a shilling, so that as long as the skin is kept clean, sweat or watery vapour is poured out from the skin. This sweat contains decaying and poisonous waste matter, which is worse than useless to the system. Hence the importance of baths and personal ablutions. Further, the healthy action of the skin bears a greater relation to our health than we might at first imagine. The function or office of the skin is closely correlated with the office of other internal organs. The skin performs the various functions of absorption, secretion, excretion, &c. If the skin fails in its duty, the work is thrown upon the liver and kidneys, consequently, they are doing double duty, and if not soon relieved they become overtaxed, and consequently injured. Further, a dirty skin affects our spirits; if the skin is dirty the terminal branches of nerves and blood vessels are not benefited and strengthened by atmospheric air. I hope the above is simple and intelligible, and that I have shown that it is impossible to enjoy life if we allow our skin to be clogged with dirt. We have seen the vital importance of fresh air and pure water. Yet, as Dr. Lankester so suggestively remarks in his address upon public health at the last Social Science Congress, "As the animal organism cannot live in air and water, but requires the varied compounds of carbon, oxygen, hydrogen, and nitrogen for its existence, the whole question of the relation of the elements used as food becomes the subject of our inquiries. Here our researches are based especially on chemical facts, and we must take for our guides Müllder and Liebig, and the great school of chemical physiologists. The freshest air and the purest water will be no protection from disease and death, unless the human system is supplied in its food with the elements necessary for the play of those chemical forces which result in life. Not only must there be food



supplying the materials of combustion and nutrition, but each tissue is built up and constituted its own peculiar way. The blood must be supplied with chloride of sodium and iron, the bones with phosphate, carbonate, and fluuate of lime; the muscles with potash; the bile with sulphur; the saliva with cyanogen; the teeth, hair and nails with silica. A diet deficient of any of these materials may be the source of disease. Our navy was formerly decimated for want of fresh vegetables, our army was starved on an excessive diet of salt beef, our children die if fed alone on arrowroot or corn-flour. Those who inhale abundance of fresh air, and have access to infinite stores of pure water, nevertheless fall easy victims to diseases which result from the rebundance or deficiency of the compounds which, in natural quantities, constitute the source of their daily life.”

From the above comprehensive sketch of the principles of diet, we see to what an extent food influences the health of the individual, and consequently the prosperity and happiness of the nation. The subject of diet is so important, so full of interest, and opens so wide a field for interesting investigation, that I feel I cannot do better than refer those who wish for scientific information conveyed in popular terms on the subject of food, to Dr. Lankester’s “Lectures on Food;” and, in conclusion, I would say that it is only when the processes of digestion, respiration, and elimination, are in proper correlation, that we can expect either to enjoy or preserve good health.

## APPENDIX.

A few brief observations on the control and prevention of infectious diseases, such as small-pox, measles, scarlet fever, typhus fever, typhoid fever, cholera, &c.

The above-mentioned diseases are capable of being transmitted from those who are infected to those who are not. It is at once cheering and obligatory to know that the spread of these infectious diseases depends upon certain specific conditions, and that these conditions are primarily within the control of man. This control and prevention of infectious diseases is one of the latest and greatest achievements of modern medicine.

Small-pox is one of the most contagious diseases, but it is controlled by vaccination. Primary vaccination within certain limits, is a protection against small-pox, but it is a well ascertained fact that there is a period in the life of persons when the prophylactic or protective power of vaccinia dies out, or at the least becomes very feeble, consequently it is advisable for all who are in the proximity of those suffering from small-pox to be re-vaccinated, for the following reasons. Re-vaccination will almost without exception restore the protective power against small-pox: secondary vaccination is attended with no danger. The operation is so simple and the protection so certain and valuable.

Scarlet fever and measles, though regarded as the inevitable diseases of childhood, and for which the mother is considered the best physician, and kitchen physic the best medicine, are simple in themselves, but on account of the sequelæ or diseases which generally follow, really require more attention in a sanitary point of view than they generally receive.

Typhus, typhoid fever, and cholera, are contagious diseases, and where the conditions of a locality are favourable they rapidly spread death and

desolation around. Typhoid fever is generally caused by a specific miasm of sewers and neglected water closets, &c.

With respect to the cause of cholera, I may mention decomposed and decomposing animal and vegetable matter in drinking water, but whatever may be the essential cause or causes of cholera, I must leave for some future discussion. What is of the most practical value to know is, that the laws which govern the endemic influence of cholera, are as well known as those of typhoid fever, and equally under our control. Those conditions which are favourable to the spread of typhoid fever, are equally favourable to the spread of cholera when once it shows itself, and the same sanitary regulations which are applicable to typhoid fever, are best calculated to prevent and control cholera.

The following suggestions may be observed with advantage to the sick, and those connected with them:—

1. In cases of contagious diseases *fresh air* is of primary importance to both patient and attendant. The room should be well ventilated, care should be taken that no draught blows upon the patient, but a continual supply of fresh air enters the room. In cases where it is not expedient to have both window and door open a fire ought to be lighted, and the door kept open. A fire not only is a good means of ventilating a sick room, but it also dries the air and thus renders it a less susceptible medium for carrying and retaining infectious poisons.

2. Cleanliness is very desirable in health, but it is imperatively necessary in disease, both the room and the patient should be kept thoroughly clean. Both body and bed linen should be frequently changed, and as soon as the soiled linen is removed it should be placed in water and thoroughly washed, and not rolled up and kept till the regular washing day. There is a great objection to change the linen of patients suffering from fever for fear they should catch cold, this is an ignorant prejudice. Fever patients do not take cold if ordinary care be employed. The skin of fever patients not only becomes hot and uncomfortable but dirty. It is often advisable to sponge them with a little luke-warm vinegar and water. The room ought to be washed out daily, and thoroughly dried. It is useful to have a little of Condry's disinfectant fluid in the water. All carpets, curtains, and woollen

furniture is better removed out of the sick chamber. All discharges from the patient ought to be immediately emptied, and the utensils washed out with Condry's fluid and water; and further, all water closets &c., ought, at these times, to be daily (at least) well flushed with water.

3. Nurses, relatives, friends, and visitors, ought to avoid breathing the patient's breath, and ought not to remain in close proximity longer than duty requires. They should not swallow their spittle, and on leaving the sick chamber, they should clear their mouth and nostrils. Smoking is only of service in so far as it promotes expectoration; and snuff, only because it induces sneezing.

With respect to disinfectants I would say, better that they should never be used, than that their use should tend to supercede fresh air and cleanliness. In order to disabuse the minds of some who look upon disinfectants as charms, I give a list of some of them, with their supposed action:—

Charcoal absorbs moisture and noxious effluvia; its value depends upon its being *fresh* and *dry*; it stops meat from becoming putrid.

Bromine is the most efficient antiseptic known, but it requires great care and caution in its employment, consequently it is best left to the direction of the medical attendant as to how, when, and where, it should be employed.

Condry's disinfectant, Darby's fluid and ozonized water, are but solutions of the permanganate of potassa, and it is upon this that their value as antiseptics and deodorizers depend. Carbolic acid and coal tar are also antiseptic and deodorant in their action. McDougall's powder contains carbolate of lime, and is a very valuable preparation in arresting putrescence and controlling effluvia. Chloride of lime is a common disinfectant, and is useful in destroying noxious compound gases.

Heat is the last disinfectant that I shall mention. It is at once cheap and effectual, and is by far the most valuable in destroying contagious virus, transportable infections. All bedding, garments, &c., should be purified by hot air or steam, as it will effectually destroy cell life, cryptogamic and infusorial organisms.



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